

Hongao ZHU

Education

Shanghai Jiao Tong University (SJTU), Shanghai, China

2021.09 – Present

B.A., English (Linguistics), Expected 2025

• Relevant Coursework:

- **Linguistics:** Semantics, Syntax, Corpus Linguistics, Phonetics and Phonology, Research Methods in Linguistics, Neurolinguistics, Language and Cognition, Language and Intelligence, Language Acquisition, Hearing Science, French, Latin
- **Math and Statistics:** Calculus, Statistics for Language and Linguistic Research, Language Data and Python Techniques
- **Computer Science:** Data Structure, Natural Language Processing, Reinforcement Learning, Machine Learning, Deep Learning

Publications

Hongao Zhu, Hai Hu. (in prep). A Comparative Study of Language Acquisition in LLMs and Children: Evidence from the *Ba* Construction in Chinese. *TACL*.

Hongao Zhu, Lei Lei. (2025). Detecting Authorship Between Generative Models and Humans: A Burrows's Delta Approach. *Digital Scholarship in the Humanities*, fqaf048.

Byung-Doh Oh, **Hongao Zhu**, William Schuler. (2025). The Inverse Scaling Effect of Pre-Trained Language Model Surprisal Is Not Due to Data Leakage. *Findings of the Association for Computational Linguistics: ACL 2025*.

Yi-Chien Lin, **Hongao Zhu**, William Schuler. (2025). Vectors from Larger Language Models Predict Human Reading Time and fMRI Data More Poorly when Dimensionality Expansion is Controlled. *ArXiv preprint*.

Hongao Zhu, Muxiaoqiao Xu, Siyuan Song, Yikang Liu, Yuxia Wang, Hai Hu. (2025). A Preliminary Analysis of the Predicative Power of Surprisal in Reading Chinese. *NAACL 2025 Workshop CMCL Submission*.

Yikang Liu, Yeting Shen, **Hongao Zhu**, Lilong Xu, Zhiheng Qian, Siyuan Song, Kejia Zhang, Jialong Tang, Pei Zhang, Baosong Yang, Rui Wang, Hai Hu. (2024). ZhoBLiMP: a Systematic Assessment of Language Models with Linguistic Minimal Pairs in Chinese. *ArXiv preprint*.

Research Experience

Assessing the Comprehension and Production in Transformer-based Large Language Models

2024.07 – Present

Undergraduate Thesis, supervised by Hai Hu

Department of Linguistics, SJTU

- Designed *Ba*-BLiMP, the first constraint-driven benchmark of the Chinese *Ba* construction and validated it using eleven LLMs.
- Trained four LMs from 14M to 410M on a carefully curated Chinese dataset featuring about 3.5B tokens and recorded their acquisition trajectories
- Transferred methods from First Language Acquisition to obtain open-ended generations from LMs and subsequently conducted error analysis.

Language Models As Surprisal Predictors to Human Reading Times

2024.03 – Present

Research Assistant, supervised by William Schuler

Department of Linguistics, Ohio State University

- Contributed to the psycholinguistic toolkit [modelblocks](#) by adding modules for deduplication and trimming human-response data.
- Deduplicated the [NatalStories](#) corpus of [Pythia](#) training data, consolidating the predictive power of small LMs proposed by [Oh and Schuler \(2023\)](#).
- Reassessed transformer-based models' alignment with humans on cognitive tasks and behaviors, following [Schrimpf et al. \(2021\)](#).

Detecting Authorship Between Generative AI Models and Humans: A Burrows's Delta Approach

2023.10 – Present

Team Leader, supervised by Lei Lei

Department of Linguistics, Shanghai International Studies University

- Appropriated the classic stylometric algorithm Burrows's Delta into AI detection and proved its robustness on the [ArguGPT](#) Corpus.
- Implemented and tested the Delta algorithm, SVM, and RNN using Python and with NLP packages such as Scikit-Learn, SciPy, and NumPy.

- Overrode the classic Delta implementations in Scikit-Learn for efficiency according to the peculiar structures of data.

ZhoBLiMP: a Systematic Assessment of Language Models with Linguistic Minimal Pairs in Chinese

2023.10 – Present

Research Assistant, supervised by Hai Hu

Department of Linguistics, SJTU

- Designed and implemented [ZhoBLiMP](#), the largest Chinese BLiMP containing 118 paradigms covering 15 linguistic phenomena.
- Trained twenty LMs of different sizes (14M–1.4B) on up to three billion Chinese tokens and evaluated them with fourteen off-the-shelf LLMs.
- Modulated ZhoBLiMP for easy scale-ups, by presenting it with a toolchain that allows customizable grammar and vocabulary.

Assessing The Communication Effect of Chinese Image in Australian Public Discourse

2022.05 – 2023.09

Team Leader, supervised by Changpeng Huan

Department of Linguistics, SJTU

- Conducted multimodal discourse, source, and strategy analyses on the self-built *Corpus of Chinese News from Australia*.
- Leveraged UCREL Semantic Analysis System and Corpus linguistics toolkits such as Wordsmith and KHCode in critical discourse analysis.

Topic Modeling And Sentiment Analysis of classical literary works across 700 years

2021.10 – 2023.09

Team Leader, supervised by Lei LEI

Department of Linguistics, SJTU

- Explored NLP techniques, topic-modeling, and sentiment analysis applied to the [Corpus of Canon of Western Literature](#).
- Invented new stop-words lists and optimized the LDA model using Perplexity calculations for historical text topic-modeling.
- Visualized analysis in interactive heat-maps and co-occurrence graphs by leveraging the assistance of NLTK, Matplotlib, and pyLDAvis.

Workshops

University of Oxford, Oxford, The United Kingdom

2023.08 – 2023.09

Attendee, Oxford Prospects Program affiliated with Regent Park's College, 2023

- Discussed the latent negativity in discourse sentiment analysis and questions about topic segmentation with Dr. Samantha-Kaye Johnston.

Hong Kong University, Hong Kong, China

2023.05 – 2023.06

Attendee, New Horizons in Digital Humanities & Cultural Data Science Workshop, 2023

- Conversated with Professor David Mimno and explored the applications of LDA and the feasibility of LLM-assisted essay scoring for TOFEL.

Shanghai International Studies University, Shanghai, China

2023.01 – 2023.02

Attendee, Workshop on Discourse and Multimodal Analysis, 2023

- Learned from lectures given by prominent professors such as David Machin about multimodal and discourse analysis.

Skills

Languages: Mandarin Chinese (Native), English (Advanced), French (Intermediate), Latin (Preliminary)

Programming Languages: Python (Scikit-learn, PyTorch, NLTK, Transformers, Pandas, Numpy...), Bash, R, LaTeX, Makefile, C/C++, C#

Software: Visual Studio Code, Overleaf, Praat, Antconc, WordSmith, [Modelblocks](#), KHCode, CLAN, PS, PR, Figma

Awards and Honors

- Excellent, Test for English Majors – Band 4 (TEM-4) 2023
- Level 3, China Accreditation Test for Translators and Interpreters (CATTI) 2023
- Level 3, National Computer Rank Examination (NCRE) 2022
- First Prize, 7th National Academic English Vocabulary Competition (NAEV) 2022
- Second Runner-up and First Prize, 2022 China International Law of the Sea Moot Court Competition 2022
- Champion, RoboCup Small Group in Shanghai Jiao Tong University 2021

Course Projects

RL Logistics Simulator, SJTU, Shanghai, China

2024.05

- Designed a smart logistics simulator using DQN and Dijkstra that can prevent 10% of the traffic jams in Black Friday events via prediction.

Acquisition of English Plural Marking by a Cantonese-English Bilingual Child: A Corpus-based study, SJTU, Shanghai, China

2023.09

- Presented a CHILDES-based study of simultaneous bilingual children to support the interdependent development hypothesis.

Extracurricular Experience

Editor and Journalist, Yiyou Newspaper, SJTU, Shanghai, China	2022.09 – 2023.09
Playwright, School History Drama for the 127 th Anniversary, SJTU, Shanghai, China	2022.02 – 2023.04
Director of the Academic Division, English Debate Club, SJTU, Shanghai, China	2021.09 – 2023.06